

Contact

Email

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LinkedIn Profile:

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Address

Choriner Straße , 10119 Berlin

Expertise

- Organic Synthesis
- Perovskite Solar Cell Fabrication
- Computational Modelling
- Python & Data analysis

Languages

English: Full professional proficiency

German: A1 (Beginner)

Academic achievements

- ANRF International Travel (ITS) Grant Recipient
- PhD Fellowship funded by the Ministry of Education, Government of India

Archana Sheshachala

Professional summary

Photovoltaics researcher specializing in perovskite solar cells with interdisciplinary expertise spanning organic synthesis, thin-film processing, and device fabrication. My research bridges molecular design and device physics by combining experimental characterization with quantum-chemical calculations to understand interfacial charge transport and improve photovoltaic efficiency and long-term stability.

Research experience:

Doctoral thesis

08/2020 - 06/2026

National Institute of Technology Karnataka (NITK), India

Thesis: Design, synthesis, and structural tailoring of heterocycle-based organic hole-transporting materials for perovskite photovoltaics.

- Developed dopant-free charge-transport materials to improve photovoltaic efficiency and operational stability in perovskite solar cells.
- Established structure-property-performance relationships linking molecular design with optoelectronic properties, bandgap tuning, film morphology, and device performance.
- Performed DFT-based computational modelling to investigate molecular structure, electronic properties, and intramolecular charge-transfer behaviour.
- Investigated device physics governing charge-carrier dynamics and recombination at the perovskite/HTM interface in PSCs.
- Led collaborative research with teams at Penn State, University of Brunei, and Toin University of Yokohama
- Guided Master's thesis students in research projects.

Master's thesis

12/2018 - 06/2019

University of Mysore, Karnataka, India

- Synthesized and characterized quinazoline-based Schiff bases and their metal complexes, followed by evaluation of their antioxidant activity.

Bachelor's project

01/2017 - 05/2017

St Joseph's University, Bengaluru, India

- Synthesized carbohydrate-based metal-organic frameworks using a green chemistry approach.

Certifications

- Idea to Innovation (INUP-i2i)- Fabrication workshop- Indian Institute of Technology Madras (IITM), India
- INUP-i2i - Hands-on training on solar cells and Fabrication - IITM, India
- Schrödinger Software certification in Molecular Modelling for Materials Science.

Conferences attended

Poster Presentations

- International Conference on Chemistry for Sustainability (ICCS) 2025 - IIT Hyderabad
- Nature Conference: Advances in Perovskite Solar Cells 2025 - Stuttgart, Germany

References

- **Prof. Udaya Kumar D**, National Institute of Technology, Karnataka, India **udayakumar@nitk.edu.in**
- **Dr Kavya S. Keremane** (Assistant Research Professor), Pennsylvania State University, University Park, PA, USA **kxk5889@psu.edu**

Education

Doctor of Philosophy in Chemistry 08/2020 - 06/2026
National Institute of Technology Karnataka (NITK), India

Masters in Chemistry 06/2017 - 06/2019
University of Mysore, Karnataka, India
Grade: 8.0 GPA

Bachelors in Science 06/2014 - 05/2017
(Physics, Chemistry, Mathematics)
St Joseph's University, Bengaluru, India
Grade: 82.94%

Skills

Experimental & Characterisation:
- Multistep organic synthesis
- Spectroscopic & analytical characterization (NMR, UV-Vis, PL, FTIR, HRMS)
- Glovebox operation and inert-atmosphere handling.
- Thin-film semiconductor fabrication and characterization, emphasizing n-i-p perovskite solar cells.

Computational Modelling & Simulation
- Molecular and periodic electronic-structure modelling (DFT): geometry optimisation, frontier orbitals, ESP and charge transfer. SCF, band structures, DOS/PDOS.
- Schrödinger Suite, Quantum ESPRESSO, MD and HPC workflows.

Programming & Data Analysis
- Python for data analysis and predictive modeling (NumPy, pandas, Matplotlib, SciPy, scikit-learn, RDKit)

Publications

- **A. Sheshachala**, K. Keremane, et al., " A Molecular-Level Exploration of Dopant-Free Pyrazine-Derived Hole Transport Materials: Investigation of Interfacial Interaction in Perovskite Photovoltaics," 2025, **ChemPlusChem (Wiley)**, 90(9), e202500124.
- **A. Sheshachala**, K. Keremane, et al., "Strategic Molecular Design and Engineering of Star-Shaped Carbazole-Triphenylamine Hybrids for Dopant-Free Hole Transport Layer in Planar Perovskite Solar Cells," 2026, **Carbon Energy (Wiley) - accepted.**
- V. G. Bhat, K. Keremane, S. K. S., **A. Sheshachala** et al., "Dopant-free hydrophobic fluorene-based hole transport materials: impact of methoxy-substituted triphenylamine and carbazole peripheral groups on the performance of perovskite solar cells," 2025, **Sustainable Energy & Fuels (RSC)** 9(10), 2769-2781.
- V. G. Bhat, S. K. S., **A. Sheshachala** et al., "A 3,5-dicyano pyridine-core spiro-free hole-transport material for perovskite solar cells," 2026, **Chemical Communications (RSC)**, 62, 17036-17040.
- S. K. S., V. G. Bhat., Kavya S. K., **A. Sheshachala** et al., "Synergistic effects of linker and peripheral groups in carbazole-based dopant-free hole-transport materials for perovskite solar cells,"2026, **Materials Advances (RSC) -accepted.**